ROMAN CHAKRABORTY

Full-Stack Developer | AI Application Developer (RAG & LLM Applications)

+91-9330586100 | rikuc08@gmail.com | Portfolio | linkedin.com/in/romanchakraborty | github.com/rimc008

PROFESSIONAL SUMMARY

Full-Stack and **AI developer** (MCA, 2026) who builds web apps with React, Next.js, Node.js, and FastAPI, and ships LLM features like hybrid RAG and multi-model pipelines using LangChain, LangGraph, and Qdrant. Skilled in REST APIs, MongoDB, MySQL, Docker, and JWT.

SKILLS

Languages: Python, JavaScript, HTML, CSS
Frontend: Next.js, React.js, Vite, Tailwind CSS, Bootstrap
Backend & Databases: Next.js (API routes), Node.js, Express.js, FastAPI, REST APIs, JWT | MongoDB, MySQL, Qdrant(Vector Database)
AI & Machine Learning: RAG, LangChain, LangGraph, Vector Embeddings, Hybrid Search, BM25, Cohere Rerank, MMR, Gemini API, Groq API
Tools: Git, GitHub, Postman, Docker, Claude Code, Render, Vercel, Cloudinary

PROJECTS

YojanaAI - AI-Powered Government Scheme Discovery Platform | GitHub | Live | June 2026 - September 2026
Next.js, FastAPI, Python, MongoDB, Qdrant, LangChain, LangGraph, Docker, REST APIs, Groq API

- Built **AI platform** that helps users find relevant Indian **government schemes** by need, eligibility, or natural-language query across **4,900+ schemes**.
- Built an **end-to-end RAG system** that embeds scheme data (**24,850+ vectors**) in Qdrant and retrieves diverse, relevant schemes with **semantic search and MMR** instead of keyword matching.
- Built a **multimodel AI pipeline** with **3 LLMs** (Gemini,Groq,OpenAI) with LangChain and LangGraph, and the full-stack platform (Next.js, FastAPI, MongoDB, Qdrant, Docker) for **eligibility filtering, scheme details, application links, and AI recommendations**.

RAgeniusAI - AI-Powered PDF Learning Assistant | GitHub | April 2026 - June 2026
React.js, Tailwind CSS , FastAPI, MongoDB, Qdrant, LangChain, Gemini API, Groq API, BM25, Cohere Rerank

- Built a **hybrid RAG** pipeline for **10 to 100+** page PDFs: Qdrant **dense search fused with BM25** (weighted), **reranked with Cohere**.
- Tested retrieval on **25+** PDFs and refined chunking for multi-column documents.
- Generated **summaries** and **notes** with **Gemini**; built **quiz and Q&A** features with **Groq**.

Davis-Avery - Full-Stack E-Commerce Platform | GitHub | September 2025 - December 2025
React.js, Node.js, Express.js, MongoDB, Razorpay, JWT, Cloudinary, Tailwind CSS

- Designed the MongoDB schema and REST API for product listings using **MVC architecture**, with category and **price filtering, search, and browsing**.
- Built a **shopping cart with size selection**, secured with **JWT authentication**, protected routes, and **rate-limited** login/signup.
- Integrated **Razorpay** in test mode with failed/pending payment handling, and built an **admin dashboard** for inventory and order management.

EDUCATION

Master of Computer Application (MCA)
Kalyani Government Engineering College, West Bengal | 2024 - 2026 | **CGPA: 8.26**
Bachelor of Science (B.Sc.) in Physics | Calcutta University, West Bengal | 2020–2023 | **CGPA: 7.87**

ACHIEVEMENTS & CERTIFICATIONS

- **Rank 82 in WBJECA** (West Bengal Joint Entrance Examination for MCA), September 2024
- **Internet of Things (IoT)** - NPTEL (SWAYAM) | Link
- **System Analysis & Design** - Kalyani Government Engineering College (KGEC) | Link